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| Experiment No. 7 |
| Apply COCOMO Model for cost estimation |
| Date of Performance: 04/09/23 |
| Date of Submission: 11/09/23 |

**Aim:** To apply COCOMO Model for cost estimation.

**Objective :** To Evaluate and Apply COCOMO MODEL for Hotel Management System for Cost Estimation

**Theory:**

Boehm proposed COCOMO (Constructive Cost Estimation Model) in 1981.COCOMO is one of the most generally used software estimation models in the world. COCOMO predicts the efforts and schedule of a software product based on the size of the software.

1. Get an initial estimate of the development effort from evaluation of thousands of delivered lines of source code (KDLOC).
2. Determine a set of 15 multiplying factors from various attributes of the project.
3. Calculate the effort estimate by multiplying the initial estimate with all the multiplying factors i.e., multiply the values in step1 and step2.

The initial estimate (also called nominal estimate) is determined by an equation of the form used in the static single variable models, using KDLOC as the measure of the size. To determine the initial effort $E_i$ in person-months the equation used is of the type is shown below

**$E_i$=a*(KDLOC)b.**

In Software Engineering, there are three stages of the COCOMO Model estimation technique are:

i. Basic COCOMO Model
ii. Intermediate COCOMO Model
iii. Complete COCOMO Model

### i. <u>Basic COCOMO Model:</u>

**Boehm** postulated that any software development project. It classifies into one of the following three categories based on the development complexity –

i. Organic
ii. Semi-detached
iii. Embedded

**Organic:**

We can classify a development project to be of the organic type if the project deals with developing a well-understood application program, the size of the development team is reasonably small, and the team members are experienced in developing similar types of projects.

**Semidetached:**

A development project classifies the semidetached type if the development team. It consists of a mixture of experienced and inexperienced staff. Team members may have limited experience in related systems but may be unfamiliar with some aspects of the system being developed.

**Embedded:**

A development project considers the embedded type, if the software being developed is strongly coupled to hardware, or if stringent regulations on the operational procedures exist. Team members may have limited experience in related systems but they may be unfamiliar with some aspects of the system being developed.

**ii. <u>Intermediate COCOMO Model:</u>**

The basic COCOMO model assumes that effort and development time are functions of the product size alone. However, a host of other project parameters besides the product size affect the effort as well as the time required to develop the product.

It uses a set of 15 cost drivers (multipliers) that determines based on various attributes of software development. These cost drivers multiply with the initial cost and effort estimates to appropriately scale those up or down.

In general, Boehm identifies the cost drivers. He also classifies as being attributes of the following items:

**Product:** The characteristics of the product that considers include the inherent complexity of the product, reliability requirements of the product, etc.

**Computer:** The Characteristics of the computer that considers include the execution speed required, storage space required, etc.
**Personnel:** The attributes of development personnel that considers include the experience level of personnel, their programming capability, analysis capability, etc.
**Development environment:** These attributes capture the development facilities available to the developers. An important parameter that considers the sophistication of the automation (CASE) tools are uses for software development.

**iii. <u>Complete COCOMO model:</u>**

A major shortcoming of both the basic and the intermediate COCOMO models is that they consider a software product as a single homogeneous entity. However, most large systems are made up of several smaller sub-systems. These sub-systems often have widely different characteristics.

The **Complete COCOMO model** considers these differences in characteristics of the subsystems and estimates the effort and development time as the sum of the estimates for the individual sub-systems.
In other words, the cost to develop each sub-system estimates separately. The complete

system cost determines as the subsystem costs. This approach reduces the margin of error in the final estimate.

**Solution:**

**COCOMO CODE:**

```
#include <iostream>
#include <cmath>
#include <tuple>

using namespace std;

tuple<float, float, float> getModeConstants(const string& mode) {
   if (mode == "organic") {
      return make_tuple(3.2, 1.05, 0.38);
   } else if (mode == "semi-detached") {
      return make_tuple(3.0, 1.12, 0.35);
   } else if (mode == "embedded") {
      return make_tuple(2.8, 1.20, 0.32);
   } else {
      return make_tuple(0.0, 0.0, 0.0); // Invalid mode, return dummy values
   }
}

void intermediate_cocomo() {
   float size;
   cout << "Enter the size of the project (in KLOC): ";
   cin >> size;

   string mode;
   cout << "Enter the mode (organic, semi-detached, embedded): ";
   cin >> mode;

   float a, b, c;
   tie(a, b, c) = getModeConstants(mode);

   if (a == 0.0) {
      cout << "Invalid mode. Please choose from 'organic', 'semi-detached', or 'embedded'." << endl;
      return;
   }
   // Calculate Effort Adjustment Factor (EAF)
   float eaf = 1.0;
   for (int i = 0; i < 15; i++) {
      float factor;
      cout << "Enter the value of cost driver " << i + 1 << " (0.9 to 1.4): ";
      cin >> factor;
      eaf *= factor;
   }

   // Calculate Organic Effort (Effort in Person-Months)
```

```
    float effort = a * pow(size, b) * eaf;

    // Calculate Development Time (in months)
    float d = 0.28 + 0.2 * (c - 0.91);
    float development_time = d * pow(effort, c);

    // Calculate Staffing
    float staffing = effort / development_time;

    // Calculate Cost (in USD)
    float labor_cost_per_month;
    cout << "Enter the average labor cost per month (in USD): ";
    cin >> labor_cost_per_month;

    float cost = effort * labor_cost_per_month;

    cout << "Estimated Effort: " << effort << " Person-Months" << endl;
    cout << "Estimated Development Time: " << development_time << " Months" << endl;
    cout << "Estimated Staffing: " << staffing << " People" << endl;
    cout << "Estimated Cost: $" << cost << " USD" << endl;
}

int main() {
    intermediate_cocomo();
    return 0;
}
```

**OUTPUT:**

```
Enter the size of the project (in KLOC): 200
Enter the mode (organic, semi-detached, embedded): organic
Enter the value of cost driver 1 (0.9 to 1.4): 1.14
Enter the value of cost driver 2 (0.9 to 1.4): 0.91
Enter the value of cost driver 3 (0.9 to 1.4): 0.85
Enter the value of cost driver 4 (0.9 to 1.4): 0.84
Enter the value of cost driver 5 (0.9 to 1.4): 1
Enter the value of cost driver 6 (0.9 to 1.4): 1
Enter the value of cost driver 7 (0.9 to 1.4): 0.9
Enter the value of cost driver 8 (0.9 to 1.4): 0.75
Enter the value of cost driver 9 (0.9 to 1.4): 1.14
Enter the value of cost driver 10 (0.9 to 1.4): 0.91
Enter the value of cost driver 11 (0.9 to 1.4): 0.91
Enter the value of cost driver 12 (0.9 to 1.4): 1
Enter the value of cost driver 13 (0.9 to 1.4): 1
Enter the value of cost driver 14 (0.9 to 1.4): 0.5
Enter the value of cost driver 15 (0.9 to 1.4): 1
Enter the average labor cost per month (in USD): 5000
Estimated Effort: 196.851 Person-Months

Estimated Development Time: 1.29515 Months
Estimated Staffing: 151.99 People
Estimated Cost: $984254 USD
```

**Conclusion:**